3.6 Localisation

Algebraically, localisation can be seen as enlargening a ring to include inverses. In terms of the ideal structure this means removing (proper) ideals which contain the newly inverted elements. Geometrically ideals correspond to points/subsets, so localisation may be viewed as reducing the set of interest.

Recall the definition of multiplicative set. Some rather canonical examples are as follows

Example 3.6.1
The set $S_f = \{1, f, f^2, ...\}$ is m.c. but not necessarily saturated. As an example consider $A = \mathbb{Z}$ and $S_n = \{1, n, n^2, ...\}$ for n composite. Then $pq \in S_n$ but $p \notin S_n$.

We denote the localisation of A at S_f by A_f (or alternatively $A[1/f]$).

Example 3.6.2
If $\mathfrak{p} \lhd A$ is a prime ideal, then $A \setminus \mathfrak{p}$ is a saturated multiplicative set. More generally, we show later that S is a saturated multiplicative set if and only if it's of the form

$$A \setminus \bigcup_i \mathfrak{p}_i$$

for some family of prime ideals. We denote the localisation of A at $\mathfrak{p}$ by $A_{\mathfrak{p}}$.

3.6.1 Rings

Definition 3.6.3 (Localisation of a ring)
Let A be a ring and S a multiplicative set. Define the set

$$S^{-1}A = \left\{ \frac{a}{s} \;\middle|\; a \in A \, s \in S \right\}$$

under the equivalence relation

$$\frac{a}{s} = \frac{b}{t} \iff u(at - bs) = 0 \quad \text{some } u \in S.$$

then this is a ring in the obvious way

Proposition 3.6.4
The set $S^{-1}A$ is a ring under the obvious ring operations. It is non-zero precisely when S is proper. There is a canonical homomorphism

$$i_S : A \to S^{-1}A$$
$$a \to \left[\frac{a}{1}\right]$$

a) $\ker(i_S) = \mathrm{Ann}(S)$
b) $S^{-1}A$ *is the zero-ring if and only if* $0 \in S$ *if and only if there exists* $s, t \in S$ *such that* $st = 0$.
c) $i_S(s)$ *is invertible for all* $s \in s$
d) i_S *is injective if and only if* S *has no zero-divisors*
e) This is an isomorphism if and only if $S \subseteq A^\star$ *already consists only of invertible elements (e.g.* $S = \{1\}$*).*

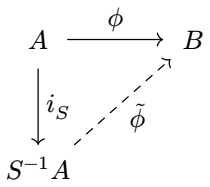
Proof.
a) This follows by the definitions
b) $\frac{1}{1} = \frac{0}{0} \iff s = 0$ for some $s \in s$ by the definitions
c) $\frac{s}{1} \cdot \frac{1}{s} = \frac{s}{s} = \frac{1}{1}$
d) This follows from the first part.
e) If $S \subseteq A^\star$ then it contains no zero-divisors and i_S is injective. Further it's clear that $\frac{a}{s} = \frac{as^{-1}}{1}$ so that the map is surjective. Similarly if the map is bijective S does not contain zero-divisors and $\frac{1}{s}$ is in the image. Therefore there is a such that $tas = 1$ for some t, which implies s is invertible.

■

Note when A is an integral domain and S is proper then the equivalence relation may be weakened to $at - bs = 0$.

Proposition 3.6.5 (Universal Property)
Let $\phi : A \to B$ be a ring homomorphism and S a multiplicative set. Then

a) There is a unique morphism $\tilde{\phi}$ making the diagram commute



if and only if $\phi(S) \subseteq B^\star$. In this case it's given by

$$\tilde{\phi}\left(\frac{a}{s}\right) = \phi(a)\phi(s)^{-1}$$

b) $\ker(\tilde{\phi}) = S^{-1}\ker(\phi)$

Proof.

a) If $\tilde{\phi}$ exists then $1 = \tilde{\phi}(1) = \tilde{\phi}(\frac{s}{1}\cdot\frac{1}{s}) = \tilde{\phi}(\frac{s}{1})\tilde{\phi}(\frac{1}{s}) = \phi(s)\tilde{\phi}(\frac{1}{s})$. Which shows that $\phi(S) \subseteq B^\star$ and $\phi(s)^{-1} = \tilde{\phi}(\frac{1}{s})$.
Conversely suppose $\phi(S) \subseteq B^\star$ then we claim that the given mapping is well-defined. For

$$\frac{a}{s} = \frac{a'}{s'} \implies s''(s'a - sa') = 0 \implies \phi(s'')\phi(s')\phi(a) = \phi(s'')\phi(s)\phi(a')$$

Multiply by the appropriate inverses to find

$$\phi(a)\phi(s)^{-1} = \phi(a')\phi(s')^{-1}$$

It's clearly a multiplicative homomorphism. Further it's additive because

$$\begin{aligned}\tilde{\phi}\left(\frac{a}{s} + \frac{b}{t}\right) &= \tilde{\phi}\left(\frac{at+bs}{st}\right)\\ &= \phi(at+bs)\phi(st)^{-1}\\ &= \phi(a)\phi(t)\phi(s)^{-1}\phi(t)^{-1} + \phi(b)\phi(s)\phi(s)^{-1}\phi(t)^{-1}\\ &= \phi(a)\phi(s)^{-1} + \phi(b)\phi(t)^{-1}\\ &= \tilde{\phi}\left(\frac{a}{s}\right) + \tilde{\phi}\left(\frac{b}{t}\right)\end{aligned}$$

b) Suppose $\tilde{\phi}(\frac{a}{s}) = 0$ then clearly $a \in \ker(\phi) \implies \frac{a}{s} \in S^{-1}\ker(\phi)$. The converse is clear.

■

In the case that A is an integral domain then generally everything becomes a lot simpler.

Example 3.6.6 (Quotient Field)
Let A be an integral domain then $A \setminus 0 = A^\star$ and we define the ***quotient field***

$$\mathrm{Frac}(A) \coloneqq (A \setminus 0)^{-1}A$$

Proposition 3.6.7 (Quotient Field contains all localisation)
Let A be an integral domain, and $\mathrm{Frac}(A)$ *the quotient field. Define another model for $S^{-1}A$ as follows*

$$S^{-1}A \coloneqq \left\{\frac{a}{s} \in \mathrm{Frac}(A) \;\middle|\; a \in A\, s \in S\right\}$$

The canonical map $A \to S^{-1}A \subset \mathrm{Frac}(A)$ is injective, and satisfies the universal property for localisation.

Proof. It's injective because A has no zero-divisors. That it satisfies the universal property is very similar as before.■

Proposition 3.6.8 (Directed Limit)
Let S_i be a family of multiplicatively closed sets directed by inclusion, such that $S = \bigcup_i S_i$ is multiplicatively closed. Then there is a canonical isomorphism

$$\varinjlim_i S_i^{-1}A \to S^{-1}A$$

induced by the canonical maps

$$S_i^{-1}A \to S^{-1}A$$

Proof. The canonical maps i_{S_iS} induce a unique morphism

$$\varinjlim_i S_i^{-1}A \to S^{-1}A$$
$$[a_i/s_i] \to a_i/s_i$$

by the universal property. An element on the right hand side is written a/s for some $s \in S$. By hypothesis $s \in S_i$ for some i, therefore it is surjective. Suppose we have two elements $[a_i/s_i]$ and $[a_j/s_j]$ on the left hand side which become equal in $S^{-1}A$. Then by definition $s_k(s_ja_i - a_js_i) = 0$ for some $s_k \in S_k$. Since it's a directed system we can find S_l containing S_i, S_j, S_k. Then by definition $a_i/s_i = a_j/s_j$ in $S_l^{-1}A$ and we see that $[a_i/s_i] = [a_j/s_j]$. Therefore the given morphism is also injective as required. ∎

Lemma 3.6.9 (Localisation Respects Contraction of Multiplicative Sets)
Let $A \subseteq B$ be subrings of a common ring C, and $T \subset B$ a multiplicative set. Writing $S \coloneqq T \cap A$ (a multiplicative set of A), and realising both localisations as subrings of C (or of a common ring of fractions), $S^{-1}A \subseteq T^{-1}B$.

In particular, for $\mathfrak{q}$ a prime ideal of B and $\mathfrak{p} \coloneqq \mathfrak{q} \cap A$, taking $T \coloneqq B \setminus \mathfrak{q}$ (so $S = A \setminus \mathfrak{p}$) gives $A_\mathfrak{p} \subseteq B_\mathfrak{q}$.

Proof. For $a/s \in S^{-1}A$ ($a \in A$, $s \in S \subseteq T$), viewing $a, s \in B$ (as $A \subseteq B$), $s \in T$ directly gives $a/s \in T^{-1}B$. ∎

Combining the previous two results, for a directed union of subrings rather than a single fixed ring:

Lemma 3.6.10 (Localisation Commutes with Directed Unions of Subrings)
Let $(A_i)_{i\in I}$ be a directed system of subrings of a common ring C, with union $A \coloneqq \bigcup_i A_i$, and $T \subset A$ a multiplicative set. Writing $T_i \coloneqq T \cap A_i$, and realising all localisations as subrings of C (or of a common ring of fractions),

$$T^{-1}A = \bigcup_i T_i^{-1}A_i$$

In particular, for $\mathfrak{p} \lhd A$ a prime ideal and $\mathfrak{p}_i \coloneqq \mathfrak{p} \cap A_i$, taking $T \coloneqq A \setminus \mathfrak{p}$ (so $T_i = A_i \setminus \mathfrak{p}_i$) gives $A_\mathfrak{p} = \bigcup_i (A_i)_{\mathfrak{p}_i}$.

Proof. If $a/s \in T^{-1}A$ ($a \in A$, $s \in T$), then $a, s \in A_i$ for some i (directed union), with $s \in T \cap A_i = T_i$, so $a/s \in T_i^{-1}A_i$. Conversely if $a/s \in T_i^{-1}A_i$ ($a, s \in A_i$, $s \in T_i \subseteq T$), then viewing $a, s \in A_i \subset A$, $s \in T$, so $a/s \in T^{-1}A$. ∎

3.6.2 Modules

Definition 3.6.11 (Localisation of a module)
Let A be a ring with S multiplicative set and M an A-module. Then we define

$$S^{-1}M = \left\{ \frac{m}{s} \;\middle|\; m \in M \right\}$$

under the obvious equivalence relation. This is then an $S^{-1}A$-module in the obvious way.

Definition 3.6.12 (Localisation of a sub-module)
Let M be an A-module and $N \subseteq M$ a sub-A-module then define

$$S^{-1}N = \left\{ \frac{n}{s} \;\middle|\; n \in M\, s \in S \right\} \subseteq S^{-1}M$$

Proposition 3.6.13
*$S^{-1}(-)$ constitutes a functor A-**Mod** $\to S^{-1}A$-**Mod**. More precisely there is a unique morphism ψ making the following diagram commute as A-module morphisms*

$$\begin{array}{ccc} N & \xrightarrow{\psi} & M \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_S} \\ S^{-1}N & \xrightarrow{S^{-1}(\psi)} & S^{-1}M \end{array}$$

where $S^{-1}(\psi)$ is in fact an $S^{-1}A$-module morphism.

It is an exact functor; for an exact sequence

$$N \to M \to P$$

the corresponding sequence of $S^{-1}A$-module morphisms

$$S^{-1}N \to S^{-1}M \to S^{-1}P$$

is exact. If N is a submodule of M then we may regard $S^{-1}N$ as a submodule of $S^{-1}M$.

Proposition 3.6.14 (Localisation commutes with quotients)
There is a commutative diagram of A-module morphisms

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N & \xrightarrow{i} & M & \xrightarrow{\pi} & M/N & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_S} & & \downarrow & & \\ 0 & \longrightarrow & S^{-1}N & \longrightarrow & S^{-1}M & \xrightarrow{S^{-1}(\pi)} & S^{-1}(M/N) & \longrightarrow & 0 \end{array}$$

with exact rows and the bottom row consists of $S^{-1}A$-module morphisms. This induces an isomorphism of $S^{-1}A$-modules.

$$S^{-1}M/S^{-1}N \cong S^{-1}(M/N)$$

Proposition 3.6.15
Suppose $N \subseteq N' \subseteq M$ then there is a canonical short-exact sequence of $S^{-1}A$-modules

$$0 \to S^{-1}(N'/N) \to S^{-1}(M/N) \to S^{-1}(M/N') \to 0$$

which induces an isomorphism

$$S^{-1}(M/N)/S^{-1}(N'/N) \cong S^{-1}(M/N')$$

Proposition 3.6.16
Let M be a finitely-generated A-module. Then

$$S^{-1}M = 0 \iff sM = 0 \text{ some } s \in S$$

3.6.3 Ideals

Recall the notion of extended and contracted ideals in Definition (3.3.48).

Definition 3.6.17 (Localisation of an ideal)
Let A be a ring and S a multiplicative set and $\mathfrak{a} \lhd A$ define

$$S^{-1}\mathfrak{a} := \left\{ \frac{a}{s} \;\middle|\; a \in \mathfrak{a} \right\}$$

then this is an ideal of $S^{-1}A$.

Proposition 3.6.18 (Extension and Contraction)
Let A be a ring with multiplicative set S and canonical morphism $i_S : A \to S^{-1}A$.

a) $\mathfrak{a}^e = i_S(\mathfrak{a})S^{-1}A = \left\{ \frac{a}{s} \mid a \in \mathfrak{a}, s \in S \right\} = S^{-1}\mathfrak{a}$

b) $\mathfrak{b}^c = \left\{ a \mid \frac{a}{1} \in \mathfrak{b} \right\}$

c) An ideal $\mathfrak{a}$ in A satisfies

$$\mathfrak{a}^{\mathrm{ec}} = \bigcup_{s \in S} (\mathfrak{a} : s) = \{ a \in A \mid as \in \mathfrak{a} \text{ some } s \in S \}$$

In particular $\mathfrak{a}$ is contracted if and only if $as \in \mathfrak{a} \wedge s \in S \implies a \in \mathfrak{a}$

d) $\mathfrak{b}$ proper $\iff \mathfrak{b}^c$ proper $\iff \mathfrak{b}^c \cap S = \varnothing$

e) $\mathfrak{a}^e$ proper $\iff \mathfrak{a} \cap S = \varnothing$

f) Every ideal $\mathfrak{b} \lhd S^{-1}A$ is extended (equiv. $\mathfrak{b} = \mathfrak{b}^{\mathrm{ce}} = S^{-1}\mathfrak{b}^c$).

g) A prime ideal $\mathfrak{p}$ is contracted if and only if $\mathfrak{p} \cap S = \varnothing$. In this case $\mathfrak{p}^e$ is prime.

h) $\mathfrak{q}$ prime $\implies \mathfrak{q}^c$ *is prime and satisfies* $\mathfrak{q}^c \cap S = \varnothing$.

Proof.

a) $S^{-1}\mathfrak{a}$ is an additive subgroup because $\frac{a_1}{s_1} + \frac{a_2}{s_2} = \frac{a_1 s_2 + a_2 s_1}{s_1 s_2}$. It contains $i_S(\mathfrak{a})$ and is closed under multiplication by A, therefore $\mathfrak{a}^e \subseteq S^{-1}\mathfrak{a}$. Similarly as $\mathfrak{a}^e$ is an ideal containing $i_S(\mathfrak{a})$, we have $\frac{a}{s} = \frac{1}{s} \cdot \frac{a}{1} \in \mathfrak{a}^e$, i.e. $S^{-1}\mathfrak{a} \subseteq \mathfrak{a}^e$ as required.

b) This is clear

c) Observe that

$$\begin{aligned}\mathfrak{a}^{\mathrm{ec}} &= \left\{ a \in A \;\middle|\; \frac{a}{1} \in \mathfrak{a}^e \right\} \\ &= \left\{ a \in A \;\middle|\; \frac{a}{1} = \frac{a'}{s} \quad a' \in \mathfrak{a}\, s \in S \right\} \\ &= \{ a \in A \mid sa \in \mathfrak{a} \text{ some } s \in S \}\end{aligned}$$

By (3.3.49) an ideal $\mathfrak{a}$ is contracted if and only if $\mathfrak{a} = \mathfrak{a}^{\mathrm{ec}}$. Furthermore it always satisfies $\mathfrak{a}^{\mathrm{ec}} \subseteq \mathfrak{a}$. The reverse inclusion is precisely the condition given.

d) This first equivalence is true in general, see (3.3.49). Clearly $\mathfrak{b}^c = A \implies \mathfrak{b}^c \cap S \neq \varnothing$. Similarly if $S \cap \mathfrak{b}^c \neq \varnothing$ then $s \in \mathfrak{b}^c \implies \frac{s}{1} \in \mathfrak{b} \implies 1 \in \mathfrak{b} \implies 1 \in \mathfrak{b}^c$.

e) By d) $\mathfrak{a}^e$ is proper if and only if $\mathfrak{a}^{\mathrm{ec}}$ is proper. By c) we see $1 \in \mathfrak{a}^{\mathrm{ec}}$ if and only if $S \cap \mathfrak{a} \neq \varnothing$ and the result follows.

f) By (3.3.49) we need only show $\mathfrak{b}^{\mathrm{ce}} \subseteq \mathfrak{b}$. Note $\frac{a}{s} \in \mathfrak{b}^{\mathrm{ce}} \implies \frac{a}{s} = \frac{a'}{s'}$ with $a' \in \mathfrak{b}^c$. By 2. $\frac{a'}{1} \in \mathfrak{b}$ and therefore so is $\frac{a}{s} = \frac{a'}{s'} = \frac{a'}{1} \cdot \frac{1}{s'} \in \mathfrak{b}$ as required.

g) If $\mathfrak{p} \cap S = \varnothing$ then by primality it automatically satisfies the conditions in c) and is therefore contracted. Conversely if a prime ideal $\mathfrak{p}$ is contracted then $\mathfrak{p} = \mathfrak{q}^c$. It is by definition proper so by d) it satisfies $\mathfrak{p} \cap S = \varnothing$ as required.

Suppose $\mathfrak{p}$ is prime and $\frac{a}{s} \cdot \frac{b}{t} \in \mathfrak{p}^e$ then $\frac{ab}{st} = \frac{x}{u}$ for $x \in \mathfrak{p} \implies v(abu - xst) = 0 \implies uvab \in \mathfrak{p} \implies a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. Therefore $\frac{a}{s} \in \mathfrak{p}^e$ or $\frac{b}{t} \in \mathfrak{p}^e$ as required.

h) Generically $\mathfrak{q}^c$ is a contracted prime ideal and we've already shown in d) that $\mathfrak{q}^c \cap S = \varnothing$.

■

Corollary 3.6.19 (Ideal Structure Localisation)
Let A be a ring and S a multiplicative set then there is an order-preserving bijection of proper ideals

$$\{\mathfrak{a} \lhd A \mid \mathfrak{a} \text{ contracted}\} \leftrightarrow \{\mathfrak{b} \lhd S^{-1}A\}$$

which restricts to a bijection of prime ideals

$$\{\mathfrak{p} \lhd A \mid \mathfrak{p} \cap S = \varnothing\} \leftrightarrow \{\mathfrak{q} \lhd S^{-1}A\}$$

Proof. From (3.6.18) every ideal of $S^{-1}A$ is extended. Therefore the bijection of proper ideals follows from (3.3.52). For prime ideals each direction is well-defined by (3.6.18) g). ■

3.6.4 Change of Rings

For what follows it is useful to have the concept of saturation of a multiplicatively closed set. Essentially taking the saturation $\overline{S}$ of S doesn't change the ring $S^{-1}A$.

Proposition 3.6.20 (Saturation)
Let A be a ring and S a multiplicatively closed set. Then the following sets are equal

a) $(i_S)^{-1}\left((S^{-1}A)^\star\right)$
b) $\{x \in A \mid ax \in S \text{ for some } a \in A\}$
c) $\bigcap_{T \supseteq S : T \text{ saturated}} T$

which we denote by $\overline{S}$. We have the following properties

- *$\overline{S}$ is saturated.*
- *S is saturated if and only if $S = \overline{S}$*
- *$\overline{\overline{S}} = \overline{S}$.*

Proof. Note $x \in (i_S)^{-1}\left((S^{-1}A)^\star\right) \implies \frac{x}{1} \cdot \frac{b}{t} = 1 \implies s(xb - t) = 0 \implies (sb)x \in S$. Similarly if $ax \in S$ then $\frac{x}{1} \cdot \frac{a}{ax} = 1$.

It's clear from b) that the set thus defined is saturated and multiplicatively closed. Let T be another saturated multiplicatively closed set containing S and suppose $ax \in S \implies ax \in T \implies x \in T$, so we find that the sets are equal.

We've proved that $\overline{S}$ is saturated. Clearly $S = \overline{S}$ implies S is saturated. Conversely if S is saturated then by c) we have $\overline{S} \subseteq S$, and clearly $S \subseteq \overline{S}$. The final part follows easily. ∎

We also give another characterization of saturated multiplicatively closed subsets

Proposition 3.6.21
Let A be a ring and S a multiplicatively closed subset. Then

$$\overline{S} = A \setminus \bigcup_{\mathfrak{p} \cap S = \varnothing} \mathfrak{p}$$

Proof. Denote the right hand side by T. Then clearly $S \subseteq T$ and as noted before in (3.6.2) T is saturated. Therefore $\overline{S} \subseteq T$ by (3.6.20) c).

Conversely suppose $a \notin \overline{S}$. Consider the principal ideal (a) then $(a) \cap S = \varnothing$ (because $ab \in S \implies a \in \overline{S}$ by (3.6.20) b)). Therefore by (3.3.40) there is a prime ideal $\mathfrak{p}$ containing a which does not intersect S. Therefore $a \notin T$. Contrapositively $T \subseteq \overline{S}$ as required. ∎

Proposition 3.6.22 (Change of Rings)
Let $\phi : A \to B$ be a ring homomorphism, S, T corresponding multiplicative subsets. Then

- *There exists a morphism $\tilde{\phi}$ making the diagram commute*

$$\begin{array}{ccc} A & \xrightarrow{\phi} & B \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_T} \\ S^{-1}A & \overset{\tilde{\phi}}{\dashrightarrow} & T^{-1}B \end{array}$$

  if and only if $\phi(S) \subseteq \overline{T}$. In this case it is unique and given by

$$\tilde{\phi}\left(\frac{a}{s}\right) = \frac{\phi(a)b'}{\phi(s)b'}$$

  where $b' \in B$ is any b' such that $\phi(s)b' \in T$.

- *If in addition $T \subseteq \phi\left(\overline{S}\right)$ then ϕ injective (resp. surjective, bijective) implies $\tilde{\phi}$ is injective (resp. surjective, bijective)*

- *Further ϕ surjective $\implies \ker\left(\tilde{\phi}\right) = S^{-1}\ker(\phi)$.*

Proof.

- If $\tilde{\phi}$ is well-defined, then $i_T(\phi(S)) = \tilde{\phi}(i_S(S)) \subseteq \tilde{\phi}\left(\left(S^{-1}A\right)^\star\right) \subseteq \left(T^{-1}B\right)^\star$, which implies $\phi(S) \subseteq i_T^{-1}\left(\left(T^{-1}B\right)^\star\right) = \overline{T}$.

  Conversely if $\phi(S) \subseteq \overline{T}$ then $(i_T \circ \phi)(S) \subseteq \left(T^{-1}B\right)^\star$ therefore by (3.6.5) the morphism exists making the diagram commute.

  Note that

$$\tilde{\phi}\left(\frac{a}{s}\right) = \tilde{\phi}(i_S(a)i_S(s)^{-1}) = \tilde{\phi}(i_S(a))\tilde{\phi}(i_S(s))^{-1}$$

  so it is uniquely defined by the commutativity condition. Note that given $s \in S$, by (3.6.20) there exists $b' \in B$ such that $\phi(s)b' \in T$. In this case it's clear that $i_T(\phi(s))^{-1} = \frac{b'}{\phi(s)b'}$ from which the explicit form results.

- Suppose $T \subseteq \phi\left(\overline{S}\right)$ and ϕ is injective. Then $\tilde{\phi}\left(\frac{a}{s}\right) = 0 \implies t\phi(a) = 0$ for $t \in T$. Then there exists $s' \in \overline{S}$ and $x \in A$ such that $xs' \in S$ and $\phi(s') = t$. Therefore $\phi(as') = 0 \implies as' = 0 \implies a(xs') = 0 \implies \frac{a}{s} = 0$ as required.

  Similarly if ϕ is surjective and given $\frac{b}{t} \in T^{-1}B$ there exists $a \in A$ such that $\phi(a) = b$ and $s \in \overline{S}$ such that $\phi(s) = t$. Then $xs \in S$, $\phi(xs) \in \overline{T}$ and $\phi(yxs) \in T$ for some $x, y \in A$. Finally

$$\tilde{\phi}\left(\frac{axy}{sxy}\right) = \frac{\phi(axy)}{\phi(sxy)} = \frac{b}{t}$$

as required.

- TODO

■

Corollary 3.6.23
Let $A \xrightarrow{\phi} B \xrightarrow{\psi} C$ be a sequence of homomorphisms and S, T, U be multiplicative sets such that $\phi(S) \subseteq \overline{T}$ and $\psi(T) \subseteq \overline{U}$, then in the notation of the previous Proposition

$$\tilde{\psi} \circ \tilde{\phi} = \widetilde{\psi \circ \phi}$$

Proof. This follows from the uniqueness condition in (3.6.22). ■

Corollary 3.6.24 (Localisation Maps)
Let A be a ring and S, T two multiplicative sets. Then TFAE

- *There exists $i_{ST} : S^{-1}A \to T^{-1}A$ such that $i_{ST} \circ i_S = i_T$*
- *$S \subseteq \overline{T}$*

In this case i_{ST} is the unique such map. We have the transitivity relationships

$$i_{TU} \circ i_{ST} = i_{SU}$$

$$i_{SS} = \mathbf{1}_{S^{-1}A}$$

and furthermore i_{ST} is an isomorphism if and only if $\overline{S} = \overline{T}$. In particular $i_{S\overline{S}}$ is an isomorphism.

Finally $\ker(i_{ST}) = S^{-1}\ker(i_T)$.

Proof. This existence of i_{ST} follows from (3.6.22) when considering the map $\phi = 1_A$. The transitivity and reflexive relationships follow from (3.6.23). ■

Corollary 3.6.25 (Localisation commutes with quotient)
Let A be a ring, $\mathfrak{a}$ an ideal and S a multiplicative set. Then there exists a unique morphism making the diagram commute

$$\begin{array}{ccc} A & \xrightarrow{\pi} & A/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_{\pi(S)}} \\ S^{-1}A & & \\ \downarrow{\scriptstyle \pi} & & \\ S^{-1}A/S^{-1}\mathfrak{a} & \xrightarrow{\sim} & \pi(S)^{-1}(A/\mathfrak{a}) \end{array}$$

which is an isomorphism, and determined by

$$\frac{a}{s} + S^{-1}\mathfrak{a} \to \frac{a + \mathfrak{a}}{s + \mathfrak{a}}$$

Note that $S \cap \mathfrak{a} \neq \varnothing \iff S^{-1}A/S^{-1}\mathfrak{a} = 0 \iff \pi(S)^{-1}(A/\mathfrak{a}) = 0$.

When $\mathfrak{b} \supseteq \mathfrak{a}$ this restricts to a commutative diagram of A-modules

$$\begin{array}{ccc} \mathfrak{b} & \xrightarrow{\pi} & \mathfrak{b}/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \downarrow{\scriptstyle i_{\pi(S)}} \\ S^{-1}\mathfrak{b} & & \\ \downarrow{\scriptstyle \pi} & & \\ S^{-1}\mathfrak{b}/S^{-1}\mathfrak{a} & \xrightarrow{\sim} & \pi(S)^{-1}(\mathfrak{b}/\mathfrak{a}) \end{array}$$

and the bottom arrow is still an isomorphism of $S^{-1}A/S^{-1}\mathfrak{a}$-modules.

Corollary 3.6.26 (Localisation commutes with quotient II)
Let A be a ring, $\mathfrak{a}$ an ideal and S a multiplicative set. Then there exists a unique morphism making the diagram commute

$$\begin{array}{ccc} A & \xrightarrow{\pi} & A/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \vdots \\ S^{-1}A & \xrightarrow{\pi} & S^{-1}A/S^{-1}\mathfrak{a} \end{array}$$

given by

$$a + \mathfrak{a} \to \frac{a}{1} + S^{-1}\mathfrak{a}$$

and it is an isomorphism precisely when every $s \in S$ is co-prime to $\mathfrak{a}$, i.e.

$$(s) + \mathfrak{a} = A \quad \forall s \in S.$$

When $\mathfrak{b} \supseteq \mathfrak{a}$ this restricts to a commutative diagram

$$\begin{array}{ccc} \mathfrak{b} & \xrightarrow{\pi} & \mathfrak{b}/\mathfrak{a} \\ \downarrow{\scriptstyle i_S} & & \vdots \\ S^{-1}\mathfrak{b} & \xrightarrow{\pi} & S^{-1}\mathfrak{b}/S^{-1}\mathfrak{a} \end{array}$$

which is an $A/\mathfrak{a}$-module morphism, and is an isomorphism when the condition (...) holds.

Proof. The condition given for being an isomorphism is equivalent to $\pi(S) \subset (A/\mathfrak{a})^\star$, which is equivalent to $i_{\pi(S)}$ being an isomorphism. Therefore we may apply (3.6.25). ∎

Corollary 3.6.27
Let A be a ring, $\mathfrak{m}$ a maximal ideal and S a multiplicative set such that $S \cap \mathfrak{m} = \varnothing$. Then there is an isomorphism

$$A/\mathfrak{m} \to S^{-1}A/S^{-1}\mathfrak{m}$$

In particular $S^{-1}\mathfrak{m}$ is a maximal ideal of $S^{-1}A$.

Proposition 3.6.28 (Transitivity)
Let $S \subset T$ be multiplicative subsets of A and let

$$i_S : A \to S^{-1}A$$

be the localisation at S. Define $T_S := i_S(T)$. Then T_S is multiplicative and there is a canonical isomorphism

$$T^{-1}A \to (T_S)^{-1}(S^{-1}A) \to \left(\overline{T_S}\right)^{-1}(S^{-1}A)$$

Furthermore if $T \subseteq U$ then $T_S \subseteq U_S$ there is a commutative diagram

$$\begin{array}{ccccc} & & S^{-1}A & & \\ & \swarrow^{i_{ST}} & \downarrow{\scriptstyle i_{T_S}} & \searrow^{i_{\overline{T_S}}} & \\ T^{-1}A & \xrightarrow{\sim} & (T_S)^{-1}(S^{-1}A) & \xrightarrow{\sim} & \left(\overline{T_S}\right)^{-1}(S^{-1}A) \\ \downarrow{\scriptstyle i_{TU}} & & \downarrow & & \downarrow \\ U^{-1}A & \xrightarrow{\sim} & (U_S)^{-1}(S^{-1}A) & \xrightarrow{\sim} & \left(\overline{U_S}\right)^{-1}(S^{-1}A) \end{array}$$

3.6.5 Algebras

Lemma 3.6.29 (Localisation is a Base Change)
Let A be a ring, S a multiplicative subset, and M an A-module (3.6.11). Then there is a natural isomorphism of $S^{-1}A$-modules

$$S^{-1}A \otimes_A M \xrightarrow{\sim} S^{-1}M, \quad \frac{a}{s} \otimes m \mapsto \frac{am}{s}.$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

The map $\Phi : S^{-1}A \times M \to S^{-1}M$, $\Phi\big(\frac{a}{s}, m\big) := \frac{am}{s}$, is well-defined: if $\frac{a}{s} = \frac{a'}{s'}$ in $S^{-1}A$ then $u(as' - a's) = 0$ in A for some $u \in S$, hence $u(as' - a's)m = 0$ in M, i.e. $u(s'(am) - s(a'm)) = 0$, i.e. $\frac{am}{s} = \frac{a'm}{s'}$ in $S^{-1}M$. Φ is clearly biadditive, and A-balanced since $\Phi\big(\frac{a}{s} \cdot b, m\big) = \frac{abm}{s} = \Phi\big(\frac{a}{s}, bm\big)$ for $b \in A$; so by the universal property of the tensor product (3.4.4) it induces a group homomorphism

$$\phi : S^{-1}A \otimes_A M \to S^{-1}M, \quad \frac{a}{s} \otimes m \mapsto \frac{am}{s},$$

which is visibly $S^{-1}A$-linear.

Conversely define $\psi : S^{-1}M \to S^{-1}A \otimes_A M$ by $\psi\big(\frac{m}{s}\big) := \frac{1}{s} \otimes m$. This is well-defined: if $\frac{m}{s} = \frac{m'}{s'}$ in $S^{-1}M$ then $u(s'm - sm') = 0$ for some $u \in S$, and then, using $\frac{1}{s} \otimes m = \frac{us'}{uss'} \otimes m = \frac{1}{uss'} \otimes us'm$ (moving $us' \in A$ across the tensor),

$$\frac{1}{s} \otimes m - \frac{1}{s'} \otimes m' = \frac{1}{uss'} \otimes us'm - \frac{1}{uss'} \otimes usm' = \frac{1}{uss'} \otimes u(s'm - sm') = 0.$$

Then $\phi\big(\psi\big(\frac{m}{s}\big)\big) = \phi\big(\frac{1}{s} \otimes m\big) = \frac{m}{s}$, so $\phi \circ \psi = \mathbf{1}$. Also $\psi\big(\phi\big(\frac{a}{s} \otimes m\big)\big) = \psi\big(\frac{am}{s}\big) = \frac{1}{s} \otimes am = \frac{a}{s} \otimes m$ (again moving a across the tensor), and since elementary tensors generate $S^{-1}A \otimes_A M$, $\psi \circ \phi = \mathbf{1}$. Hence ϕ is an isomorphism, with inverse ψ. ∎

Corollary 3.6.30 (Localisation of an Algebra is a Base Change)
Let A be a ring, S a multiplicative subset, and B an A-algebra. Then $S^{-1}B$ is naturally an $S^{-1}A$-algebra (via $S^{-1}A \to S^{-1}B$, $\frac{a}{s} \mapsto \frac{ab_0}{s}$ where $b_0 \in B$ is the image of 1_A, the unique extension of $A \to B \to S^{-1}B$ along i_S (3.6.5), as $i_S(S) \subset (S^{-1}B)^\star$), and the isomorphism of $S^{-1}A$-modules furnished by (3.6.29) (with $M := B$),

$$S^{-1}A \otimes_A B \xrightarrow{\sim} S^{-1}B, \quad \frac{a}{s} \otimes b \mapsto \frac{ab}{s},$$

is moreover an isomorphism of rings, where $S^{-1}A \otimes_A B$ carries the ring structure of (3.4.34).

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write ϕ for the $S^{-1}A$-module isomorphism of (3.6.29). It remains only to check ϕ is multiplicative on elementary tensors, since it is already additive:

$$\phi\left(\left(\frac{a}{s} \otimes b\right) \cdot \left(\frac{a'}{s'} \otimes b'\right)\right) = \phi\left(\frac{aa'}{ss'} \otimes bb'\right) = \frac{aa'bb'}{ss'} = \frac{a}{s} \cdot b \cdot \frac{a'}{s'} \cdot b' = \phi\left(\frac{a}{s} \otimes b\right) \cdot \phi\left(\frac{a'}{s'} \otimes b'\right).$$

As every element of $S^{-1}A \otimes_A B$ is a sum of elementary tensors, and ϕ is additive, ϕ is a ring homomorphism; being also a bijection, it is a ring isomorphism. ∎

Corollary 3.6.31 (Localisation of a Domain is a Base Change to the Fraction Field)
Let $A \subseteq B$ be an inclusion of integral domains, $K := \mathrm{Frac}(A)$, and $S := A \setminus \{0\}$, so that $S^{-1}B$ is naturally a subring of $\mathrm{Frac}(B)$ (as $S \subset B \setminus \{0\}$ too). Then there is a natural isomorphism of rings

$$B \otimes_A K \xrightarrow{\sim} S^{-1}B, \quad b \otimes x \mapsto bx,$$

where the right-hand side is computed in $\mathrm{Frac}(B)$, regarding $B, K \subseteq \mathrm{Frac}(B)$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

By (3.6.30) (applied to the A-algebra B, noting $S^{-1}A = K$) there is a ring isomorphism $K \otimes_A B \xrightarrow{\sim} S^{-1}B$, $\frac{a}{s} \otimes b \mapsto \frac{ab}{s}$. Composing with the symmetry isomorphism $B \otimes_A K \cong K \otimes_A B$, $b \otimes x \mapsto x \otimes b$ (3.4.7) (which is also a ring isomorphism, since swapping factors clearly preserves the componentwise multiplication of (3.4.34)), yields a ring isomorphism $B \otimes_A K \xrightarrow{\sim} S^{-1}B$ sending $b \otimes \frac{a}{s} \mapsto \frac{ab}{s}$; taking $a = 1$ this is $b \otimes x \mapsto x \cdot b = bx$ for $x = \frac{1}{s}$, and since $x = \frac{a}{s} = a \cdot \frac{1}{s}$ for general $x \in K$, bilinearity gives $b \otimes x = ab \otimes \frac{1}{s} \mapsto \frac{ab}{s} = (ab) \cdot \frac{1}{s} = b \cdot \big(a \cdot \frac{1}{s}\big) = bx$ on every elementary tensor.

Finally $\frac{ab}{s} \in S^{-1}B \subseteq \mathrm{Frac}(B)$, computed there, agrees with the product bx computed in $\mathrm{Frac}(B)$ (as $B, K \subseteq \mathrm{Frac}(B)$). ∎

Remark 3.6.32
(3.6.30) recovers (3.6.41) (relative localisation at a prime) as the special case $S := A \setminus \mathfrak{p}$ of a general ring homomor-

phism $\phi : A \to B$: the module localisation $S^{-1}B$ (with B regarded as an A-module via ϕ) is exactly $\phi(S)^{-1}B = B_{\mathfrak{p}}$, since the two defining equivalence relations on fractions agree termwise.

3.6.6 Localisation at an element

Definition 3.6.33 (Localisation at an element)
Let A be a ring and $f \in A$. Then define

$$S_f = \{1, f, ..., f^n, ...\}$$

and

$$A_f := \left(S_f\right)^{-1} A$$

We have canonical maps

$$i_f : A \to A_f$$

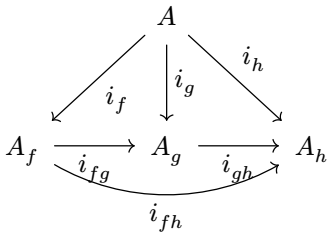
given by $i_f := i_{S_f}$.

Note that A_f is a zero ring iff f is nilpotent.

Proposition 3.6.34 (Transition maps for localisation at an element)
Let A be a ring and $f, g \in A$ then

$$S_f \subseteq \overline{S_g} \iff f \mid g^N \text{ some N} > 0$$

in this case define $i_{fg} = i_{S_f S_g}$ to be the unique morphism such that $i_{fg} \circ i_f = i_g$ (3.6.24). In addition if $h \in A$ and $S_g \subseteq \overline{S_h}$ we have a commutative diagram

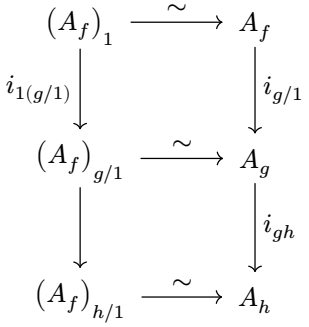


Furthermore $\overline{S_1} = A^\star$ and i_1 is an isomorphism.

Proposition 3.6.35 (Transitivity of localizing at elements)
Let A be a ring and $f, g \in A$ such that $S_f \subseteq \overline{S_g}$. There is a canonical isomorphism

$$A_g \xrightarrow{\sim} \left(A_f\right)_{g/1}$$

Furthermore $S_g \subseteq \overline{S_h} \implies S_{g/1} \subseteq \overline{S_{h/1}}$ and there is a commutative diagram



with the horizontal arrows isomorphisms and the vertical arrows are well-defined.

Proof. The existence of the isomorphism is from (3.6.28) as $i_f(S_g) = S_{g/1}$. The second statement follows because $g \mid h^N \implies g/1 \mid h^N/1$ trivially. ∎

Proposition 3.6.36
Let A be a ring and $f \in A$. Then localisation map $A \to A_f$ induces an order-preserving correspondence of prime ideals

$$\{\mathfrak{p} \lhd A \mid f \notin \mathfrak{p}\} \leftrightarrow \{\mathfrak{p} \lhd A_f\}$$

Proof. The correspondence of prime ideals follows from (3.6.19) once we observe that $S_f \cap \mathfrak{p} = \varnothing \iff f \notin \mathfrak{p}$. ∎

Proposition 3.6.37
Let A be a ring, S a multiplicatively closed subset and $g \in S$. Then the canonical map $A_g \to S^{-1}A$ has kernel $\ker(i_S)_g$. In particular when $\ker(i_S)$ is finitely generated it is injective iff $g^r \ker(i_S) = 0$ for some $r > 0$.

When A is Noetherian then there always exists g such $A_g \to S^{-1}A$ is injective, and any multiple of g also satisfies this property.

Proof. The first statement follows from (3.6.4).

When A is Noetherian then $\ker(i_S) = (a_1, ..., a_n)$ for some $a_i \in A$. By definition there exists some s_i such that $s_i a_i = 0$, whence we may take $g = s_1 \cdots s_n$. ∎

Proposition 3.6.38
Let A be a ring, S a multiplicatively closed subset and $M_1, ..., M_n$ a family of A-modules. Then there is an isomorphism

$$S^{-1}\left(\bigoplus_{i=1}^{n} M_i\right) \xrightarrow{\sim} \bigoplus_{i=1}^{n} S^{-1}M_i$$

3.6.7 Localisation at a prime ideal

Definition 3.6.39 (Localisation at a prime ideal)
Let A be a ring and $\mathfrak{p} \lhd A$ a prime ideal. Then $S := A \setminus \mathfrak{p}$ is a saturated multiplicatively closed subset, and we define

$$A_{\mathfrak{p}} := (A \setminus \mathfrak{p})^{-1}A$$

For an ideal $\mathfrak{a} \lhd A$ write the extended ideal

$$\mathfrak{a}A_{\mathfrak{p}} := S^{-1}\mathfrak{a} = \mathfrak{a}^e.$$

Definition 3.6.40 (Relative localisation at a prime ideal)
Let $\phi : A \to B$ be a ring homomorphism and $\mathfrak{p} \lhd A$ a prime ideal. Define

$$B_{\mathfrak{p}} := \phi(A \setminus \mathfrak{p})^{-1}B$$

For an ideal $\mathfrak{a} \lhd A$ write

$$\mathfrak{a}B_{\mathfrak{p}} := (\phi(\mathfrak{a})B)B_{\mathfrak{p}}$$

Observe $B_{\mathfrak{p}} = 0 \iff \mathfrak{p} \subsetneq \ker(\phi)$, so we would typically assume $\ker(\phi) \subseteq \mathfrak{p}$.

Proposition 3.6.41 (Relative Localisation as a Tensor Product)
Let $\phi : A \to B$ be a ring homomorphism (so B is an A-algebra) and $\mathfrak{p} \lhd A$ a prime ideal, $S := A \setminus \mathfrak{p}$. Then $B_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-algebra via $\frac{a}{s} \mapsto \phi(a)/\phi(s)$ (well-defined as $\phi(S) \subseteq B_{\mathfrak{p}}^{\star}$ already), and there is an isomorphism of $A_{\mathfrak{p}}$-algebras

$$\Psi : B \otimes_A A_{\mathfrak{p}} \xrightarrow{\sim} B_{\mathfrak{p}}, \quad b \otimes \frac{a}{s} \mapsto \frac{\phi(a)b}{\phi(s)},$$

with inverse $\Theta(b/\phi(s)) := b \otimes \left(\frac{1}{s}\right)$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Ψ is well-defined. For fixed $\frac{a}{s} \in A_{\mathfrak{p}}$, $b \mapsto \phi(a)b/\phi(s)$ is A-linear in b ($B_{\mathfrak{p}}$ being an A-module via ϕ); for fixed $b \in B$, if $\frac{a}{s} = \frac{a'}{s'}$ in $A_{\mathfrak{p}}$ then $s''(as' - a's) = 0$ for some $s'' \in S$, so applying ϕ and multiplying by b, $\phi(s'')(\phi(a)\phi(s') - \phi(a')\phi(s))b = 0$ in B, giving $\phi(a)b/\phi(s) = \phi(a')b/\phi(s')$ in $B_{\mathfrak{p}}$; and $\frac{a}{s} \mapsto \phi(a)b/\phi(s)$ is A-linear in $\frac{a}{s}$ (direct check, as in (3.6.5)). So the map $B \times A_{\mathfrak{p}} \to B_{\mathfrak{p}}$, $\left(b, \frac{a}{s}\right) \mapsto \phi(a)b/\phi(s)$, is A-bilinear, and induces Ψ by the universal property of the tensor product (3.4.4).

Θ is well-defined. Write $a \cdot b := \phi(a)b$ for the A-module structure on B. If $b/\phi(s) = b'/\phi(s')$ in $B_{\mathfrak{p}}$ then $s'' \cdot (s' \cdot b) = s'' \cdot (s \cdot b')$ in B for some $s'' \in S$; using $(a \cdot b) \otimes x = b \otimes (ax)$ (the defining relation of $\otimes_A$) twice,

$$b \otimes \frac{1}{s} = (s''s') \cdot b \otimes \frac{1}{s''s's} = (s'' \cdot (s' \cdot b)) \otimes \frac{1}{s''s's} = (s'' \cdot (s \cdot b')) \otimes \frac{1}{s''ss'} = b' \otimes \frac{1}{s'},$$

so Θ is well-defined on $B_{\mathfrak{p}}$.

Θ and Ψ are mutually inverse. $\Psi\Theta(b/\phi(s)) = \Psi\left(b \otimes \frac{1}{s}\right) = \phi(1)b/\phi(s) = b/\phi(s)$. And $\Theta\Psi\left(b \otimes \frac{a}{s}\right) = \Theta(\phi(a)b/\phi(s)) = (\phi(a)b) \otimes \left(\frac{1}{s}\right) = b \otimes \left(\frac{a}{s}\right)$ (the same tensor relation as above), so $\Theta\Psi$ is the identity on simple tensors, hence (by linearity) on all of $B \otimes_A A_\mathfrak{p}$.

Ψ is $A_\mathfrak{p}$-linear. For $\frac{a'}{s'} \in A_\mathfrak{p}$,

$$\Psi\left(\frac{a'}{s'} \cdot \left(b \otimes \frac{a}{s}\right)\right) = \Psi\left(b \otimes \frac{aa'}{ss'}\right) = \frac{\phi(aa')b}{\phi(ss')} = \frac{\phi(a')}{\phi(s')} \cdot \frac{\phi(a)b}{\phi(s)} = \frac{a'}{s'} \cdot \Psi\left(b \otimes \frac{a}{s}\right),$$

using the $A_\mathfrak{p}$-algebra structure on $B_\mathfrak{p}$ displayed above; so Ψ is $A_\mathfrak{p}$-linear on simple tensors, hence (by linearity) on all of $B \otimes_A A_\mathfrak{p}$.

Multiplicativity. By the algebra tensor product structure on $B \otimes_A A_\mathfrak{p}$, $\left(b \otimes \frac{a}{s}\right)\left(b' \otimes \frac{a'}{s'}\right) = bb' \otimes \frac{aa'}{ss'}$, and

$$\Psi\left(\left(b \otimes \frac{a}{s}\right)\left(b' \otimes \frac{a'}{s'}\right)\right) = \frac{\phi(aa')bb'}{\phi(ss')} = \frac{\phi(a)b}{\phi(s)} \cdot \frac{\phi(a')b'}{\phi(s')} = \Psi\left(b \otimes \frac{a}{s}\right)\Psi\left(b' \otimes \frac{a'}{s'}\right),$$

so Ψ is multiplicative on simple tensors, hence (by linearity) a ring homomorphism – and so, with $A_\mathfrak{p}$-linearity above, an isomorphism of $A_\mathfrak{p}$-algebras. ∎

Proposition 3.6.42
Let A be a ring and $\mathfrak{p}$ a prime ideal. Consider the localisation $A \to A_\mathfrak{p}$. Then there is an order-preserving bijection between (prime) ideals contained in $\mathfrak{p}$ and (prime) ideals of $A_\mathfrak{p}$

$$\{\mathfrak{q} \lhd A \mid \mathfrak{q} \subseteq \mathfrak{p}\} \leftrightarrow \{\mathfrak{q} \lhd A_\mathfrak{p}\}$$
$$\mathfrak{q} \to \mathfrak{q}A_\mathfrak{p}$$

In particular $A_\mathfrak{p}$ is a local ring with unique maximal ideal $\mathfrak{p}A_\mathfrak{p}$.

Proof. Clearly $\mathfrak{q} \cap (A \setminus \mathfrak{p}) = \varnothing \iff \mathfrak{q} \subseteq \mathfrak{p}$, so the result follows from (3.6.19). ∎

Proposition 3.6.43
Let A be a ring and $\mathfrak{p}$ a prime ideal. Then the map

$$\mathrm{Frac}(A/\mathfrak{p}) \xrightarrow{\sim} A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$$
$$\frac{\overline{a}}{\overline{b}} \to \frac{\overline{a/1}}{\overline{b/1}} = \overline{\left(\frac{a}{b}\right)}$$

is a ring isomorphism

Proof. Consider the ring map $A \to A_\mathfrak{p} \to A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$. Then this has kernel $\mathfrak{p}^{\mathrm{ec}} = \mathfrak{p}$ (3.6.18), so may be factored as an injective map $A/\mathfrak{p} \hookrightarrow A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$. By (3.6.42) $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ is a field so by universal property of localisation (3.6.5) this extends to an injective map $\mathrm{Frac}(A/\mathfrak{p}) \hookrightarrow A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ of the given form. We may verify directly that it is surjective. ∎

Proposition 3.6.44 (Localisation at prime is direct limit of localisation at an element)
Let A be a ring and $\mathfrak{p}$ a prime ideal then

$$S_\mathfrak{p} := A \setminus \mathfrak{p} = \bigcup_{f \in A \setminus \mathfrak{p}} \overline{S_f}$$

Therefore there are canonical morphisms (for $f \notin \mathfrak{p}$)

$$i_{S_f S_\mathfrak{p}} : A_f \to A_\mathfrak{p}$$

Furthermore the family of multiplicatively closed sets $\{S_f\}_{f \notin \mathfrak{p}}$ (resp. rings $\{A_f\}_{f \notin \mathfrak{p}}$) form a directed system under the relation $S \prec T \iff S \subseteq \overline{T}$. Therefore we have a canonical ring homomorphism

$$\varinjlim_{f \notin \mathfrak{p}} A_f \to A_\mathfrak{p}$$

which is an isomorphism.

Proof. As $A \setminus \mathfrak{p}$ is a saturated multiplicatively closed set we have $f \in A \setminus \mathfrak{p} \iff S_f \subseteq A \setminus \mathfrak{p} \iff \overline{S_f} \subseteq A \setminus \mathfrak{p}$. Therefore the expression for $S_\mathfrak{p}$ follows.

The family of multiplicatively closed subsets is a directed system because $S_f \subseteq \overline{S_{fg}}$. To see this note $fg \in \overline{S_{fg}} \implies f \in \overline{S_{fg}} \implies S_f \subseteq \overline{S_{fg}}$.

The final isomorphism follows because we can decompose it into two maps

$$\varinjlim_{f\notin\mathfrak{p}} A_f \xrightarrow{\sim} \varinjlim_{f\notin\mathfrak{p}} \overline{S_f}^{-1}A \xrightarrow{\sim} A_{\mathfrak{p}}$$

The first is an isomorphism by (3.6.24) and the second by (3.6.8), in light of the first statement. ∎

Proposition 3.6.45 (Quotient commutes with Localisation at Prime ideal)
Let A be a ring, $\mathfrak{a}$ an ideal and $\mathfrak{p}\supset\mathfrak{a}$ a prime ideal. Then there is an isomorphism of non-zero rings

$$A_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}} \cong (A/\mathfrak{a})_{\mathfrak{p}/\mathfrak{a}}$$
$$\frac{a}{s}+\mathfrak{a}_{\mathfrak{p}} \to \frac{a+\mathfrak{a}}{s+\mathfrak{a}}$$

Proof. Let $\pi: A\to A/\mathfrak{a}$ be the quotient map then we claim $\pi(A\setminus\mathfrak{p})=\pi(A)\setminus\pi(\mathfrak{p})$. As π is surjective we only need to show that $\pi^{-1}(\pi(\mathfrak{p}))=\mathfrak{p}$. That is if $\pi(a)=\pi(b)$ and $b\in\mathfrak{p}$ then $a\in\mathfrak{p}$. However this follows by assumption because $\mathfrak{a}\subset\mathfrak{p}$. Therefore we may apply (3.6.25). ∎

Proposition 3.6.46 (Minimal Primes consist of Zero Divisors)
Let $\mathfrak{p}$ be a proper prime ideal minimal over $\mathfrak{a}$. Then for every $x\in\mathfrak{p}$ there exists $a\in A\setminus\mathfrak{a}$ such that $xa\in\mathfrak{a}$.

In particular a minimal prime ideal consists of zero-divisors.

Proof. Observe that $\mathfrak{p}/\mathfrak{a}$ is a proper minimal prime ideal of $A/\mathfrak{a}$ by (3.3.55). Then the first statement is equivalent to showing every $x\in\mathfrak{p}/\mathfrak{a}$ is a zero divisor and we may reduce to the case $\mathfrak{a}=(0)$.

Consider the ring $A_{\mathfrak{p}}$, it has a unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ which is also minimal by (3.6.42). Every other prime ideal is contained in $\mathfrak{p}A_{\mathfrak{p}}$ by (3.3.36), and therefore equal by minimality. Therefore $\sqrt{(0)A_{\mathfrak{p}}}=\mathfrak{p}A_{\mathfrak{p}}$ by (3.3.45). Suppose $x\in\mathfrak{p}$ then $\frac{x^n}{1}=0$ and $tx^n\in\mathfrak{a}$ for $t\notin\mathfrak{p}$. Choose n minimal subject to this condition. Then $n\geq 1$ because $t\notin\mathfrak{a}$, and so $a=tx^{n-1}$ has the required properties. ∎

3.6.8 Finiteness Results

Lemma 3.6.47
Let M be an A-module, N a submodule and $f_1,...,f_r\in A$ such that

a) $N_{f_i}\to M_{f_i}$ is surjective for all $i=1...r$
b) $A=(f_1,...,f_r)$ as an ideal

then $M=N$.

Proof. Consider the A-module $P:=M/N$. Suppose $P\neq 0$, then $0\neq p\in P$ and $\mathrm{Ann}(p)$ is a proper ideal of A. In particular $f_i\notin\mathrm{Ann}(p)$ for some i. This shows $P_{f_i}\neq 0$. From (3.6.14) we have $P_{f_i}\xrightarrow{\sim} M_{f_i}/N_{f_i}$ and so $N_{f_i}\neq M_{f_i}$ a contradiction. ∎

Proposition 3.6.48
Let M be an A-module. Then

a) M finitely-generated $\implies M_f$ is a finitely generated A_f-module for all $f\in A$
b) $(f_1,...,f_r)=A$ and M_{f_i} a finitely generated A_{f_i}-module for all $i=1...r\implies M$ is a finitely-generated A-module

Proof.
a) Let $m_1,...m_n\in M$ be a generating set. Then it is easy to verify that $m_1/1,...,m_n/1$ is a generating set for M_f.
b) For each $i=1...r$, let m_{ij} be such that $m_{ij}/1$ generates M_{f_i} as an A_{f_i}-module. Let N be the A-submodule of M generated by the m_{ij}. Then $N_{f_i}=M_{f_i}$. We conclude by (3.6.47) that $N=M$ is finitely generated as an A-module.

∎